two. As a result, the higher voltage battery will discharge its current into the other battery until both are at equal voltage value.

A voltaic cell is an electrochemical cell that uses a chemical reaction to produce electrical energy. In this cell, Anode is an electrode where oxidation occurs. Cathode is an electrode where reduction occurs.

In a voltaic cell, copper and zinc are the two electrodes and sulphuric acid is an electrolyte. When placed together, a chemical reaction occurs between the electrodes and the sulphuric acid.

This reaction produces a negative charge on the zinc (surplus of electron) and a positive charge on the copper (deficiency of electrons). If an external circuit is connected across the two electrodes, electrons will flow from the negative zinc electrode to the positive copper electrodes as shown in fig 4.5.

The electric current will flow as long as the chemical action continues. In this type of cell, the zinc electrode is eventually consumed as a part of the chemical reaction.

The voltaic cell is also known as wet cell, because in this, a liquid solution is used as an electrolyte.

We learn that most of the primary cell except rechargeable ones are used one time only. It does not supply current continuously. The secondary cells overcome this disadvantage.

4.3 VOLTAIC CELL

In a secondary cell, the charging and discharging processes are taking place according to Faraday's law of electrolysis.

4.4 SECONDARY CELL

A cell that can be recharged by sending electric current in the reverse direction to that of a discharge mode is known as a secondary cell. Secondary cells are Storage batteries since, after it is charged, it stores the energy until it is used or discharged.

4.4.1 Secondary Cell Classification

Secondary cells may be classified as

(i) Lead acid cell
(ii) Alkaline cell

Example: Nickel iron cell, Nickel cadmium cell

Secondary cell is a type of electrical battery, which can be charged, discharged into a load, and recharged many times, as opposed to a disposable or primary battery, which is supplied fully charged and discharged after use. It is composed of one or more electro chemical cells. The term 'accumulator' is used, as it accumulates and stores energy through a reversible electrochemical reaction. Rechargeable batteries are produced in many different shapes and sizes, ranging from button cells to mega watt systems connected to stabilize an electrical distribution network.

Several different combinations of electrode materials and electrolytes are used, including Lead-acid, Nickel-cadmium, Nickel-metal hydride, and Lithium ion.

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